

From outcrops of rock exposed to the sea to desert areas where the only place they can grow is inside the rock itself, lichens survive in the harshest parts of the world. They are some of nature's pioneers, creating foundations on which others can live.

A detailed cross-section of a coral reef. The image shows a variety of coral structures, including branching corals and more massive, table-like corals. A clownfish is visible swimming in the water column above the reef. The water is depicted with green and blue wavy lines, suggesting movement and depth. The overall scene is a vibrant representation of a healthy marine ecosystem.

DEBATE

Lichen evolution remains a mystery. Scientists are still trying to understand how and why fungi and algae came to live together. It may have started as an attack on one by the other, which developed into a partnership. Not all lichen associations are necessarily mutually beneficial – some may be parasitic rather than a partnership.

Lichens broadly fall into three types, based on shape: foliose lichens, which have leaves; crustose lichens, which form a crust; and fruticose lichens, which have branches. However, there are some that defy this categorization, such as filamentous (hair-like) and gelatinous (water-absorbent) lichens.

Many lichens reproduce sexually by means of spores. These are produced by the fungal half of the partnership and are usually formed in special cup- or disc-like structures called apothecia. These spores, once ejected, must land next to a suitable algal partner if they are to form another lichen and survive. Other lichens produce spores inside special chambers called perithecia, which are like microscopic volcanoes, releasing their spores through a hole at the top. Alternatively, lichens may reproduce asexually by budding or breaking off specialized parts of their body. These soredia or isidia contain a mix of fungal and algal cells, which, if they land in a suitable habitat, will go on to form new lichen colonies. Rocky shorelines in North America are home to vast colonies of lichens several kilometres in length. These will have taken hundreds or even thousands of years to spread to such a huge size.

Typical of lichens, *Rhizocarpon* is able to colonize harsh environments such as dry, exposed rock surfaces.

